OOS DEVIATION & QUALITY INCIDENT INVESTIGATION REPORT

STATUS: **OPEN / UNDER RCA**
SEVERITY: **CRITICAL (OOS)**

SECTION 1: INCIDENT METADATA & INTAKE ORIGIN

Deviation Tracking ID:	**DEV-2026-0882**	**Initiation Date:**	2026-10-18	**Discovery Date:**	2026-10-19
Lead Investigator:	Dr. Arthur Vance (QA Lead)	**Department:**	Quality Assurance / QC API	**Intake Source:**	OOS Laboratory Notification

SECTION 2: PRODUCT & API MANUFACTURING CONTEXT

Active Substance:	**Artemicort Micronized Bulk API (Intermediate)**	**Batch / Lot ID:**	**ART-2026-042**
Manufacturing Unit:	Unit 3 Chemical Synthesis Cleanroom Bay 4	**Equipment Tag:**	Crystallizer Reactor CR-04 (2,500 L)
Unit Operation:	Stage 3 Controlled Anti-Solvent Crystallization	**Affected Quantity:**	480.0 kg (16 fiber drums with dual LDPE)

SECTION 3: OUT-OF-SPECIFICATION (OOS) ANALYTICAL FINDINGS

Tested Parameter / Method	Registered Specification	Observed Lab Result	Delta / Disposition
Unspecified Impurity at RRT 1.18 HPLC Assay Method STM-ART-204 Rev 03	**NMT 0.10% (w/w)** Standard Release Spec SPC-ART-004	**0.28% (w/w)** **Confirmatory re-test: 0.27%**	**+0.18% EXCEEDANCE** **CONFIRMED OUT-OF-SPECIFICATION**
Total Related Substances HPLC Sum of Impurities	**NMT 0.50% (w/w)** Standard Release Spec SPC-ART-004	**0.62% (w/w)** **Individual + Related**	**+0.12% EXCEEDANCE** **OOS — REJECT RECOMMENDED**

SECTION 4: CHRONOLOGICAL TIMELINE & PROCESS ANOMALY DESCRIPTION

Incident Narrative: On 2026-10-18 at 14:15 IST, Quality Control released finished testing results for bulk batch ART-2026-042. HPLC chromatographic assay identified an elevated unspecified peak at Relative Retention Time (RRT) 1.18 measuring 0.28% (w/w), exceeding the registered release limit of 0.10%. Total impurities also breached specification at 0.62% (limit <= 0.50%). Phase I laboratory investigation ruled out analytical instrumentation failure, column deterioration, or weighing errors through double independent injection.

Manufacturing Deviation Sequence: Cross-referencing batch production records (BPR-ART-042) revealed that during Stage 3 crystallization at 02:40 IST on 2026-10-17, secondary chiller unit CH-02 experienced an intermittent flow valve alarm (AL-CH-309). Coolant flow rate into the CR-04 reactor jacket plummeted from 45.0 L/min to 8.2 L/min. As a consequence, reactor temperature plateaued at 44.5°C for approximately 85 minutes, interrupting the programmed cooling curve (68.0°C to 12.0°C over 6.0 hours). Extended thermal exposure under agitation (65 RPM) at 44.5°C provided sufficient activation energy to drive dimerization into the RRT 1.18 degradation product.

SECTION 5: IMMEDIATE CONTAINMENT ACTIONS EXECUTED

#	Containment Action Item	Responsible Role	Verification / Status
1	SAP Quality Quarantine status applied (Block Code Q-400). Electronic transfer locked.	QA Materials Officer	**COMPLETED (SAP ID #9921)**
2	All 16 fiber drums physically quarantined in Warehouse Bay 3B with serialized red security tags (Tags #Q-042-01 through 16).	Warehouse Lead	**COMPLETED (Verified Bay 3B)**

3	Crystallizer CR-04 tagged out of service (LOTO Tag #CR04-2026-88) pending mechanical chiller inspection.	Maintenance Supervisor	**COMPLETED (LOTO Active)**

SECTION 6: UNRESOLVED EVIDENCE GAPS & PENDING AUDIT RECORDS

Record Category	Specific Missing Evidence / Unresolved Question	Investigation Impact
SCADA Historian	Continuous 1-second temperature and agitator motor torque logs for Crystallizer CR-04 between 01:00 and 06:00 IST pending retrieval from archive server.	Crucial to verify exact thermal plateau duration and cooling kinetics.
Metrology Audit	Dual-RTD Sensor TT-204 calibration certificate and Coolant Flow Transmitter FT-309 drift history pending review.	Required to rule out thermal sensing drift during crystallization.
Raw Material CoA	Supplier Certificate of Analysis for Reagent Grade Isopropanol (Lot IPA-8839) not yet verified for trace peroxides.	Ensures solvent oxidation impurities did not seed the RRT 1.18 degradation.
Operator Logbook	Night Shift B operator logbook entry regarding technician response time to Alarm AL-CH-309.	Determines why manual override was delayed by 85 minutes.

Prepared by (Lead Investigator): Dr. Arthur Vance, QA Senior Specialist Date: 2026-10-19 | Emp ID: QA-4091	**Reviewed by (QC Lab Director):** Dr. Elena Rostova, VP Analytical Science Date: 2026-10-19 | Emp ID: QC-1022	**Concurrence (Manufacturing Head):** Marcus Thorne, P.Eng., Operations Director Date: 2026-10-19 | Emp ID: PR-8823

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